
NAME

RDKitGenerateMolecularFrameworks.py - Generate Bemis Murcko molecular frameworks

SYNOPSIS

```
RDKitGenerateMolecularFrameworks.py [--infileParams <Name,Value,...>] [--mode <GraphFrameworks or AtomicFrameworks>] [--outfileParams <Name,Value,...>] [--overwrite] [--removeDuplicates <yes or no>] [--sort <yes or no>] [--sortOrder <ascending or descending>] [--useChirality <yes or no>] [-w <dir>] -i <infile> -o <outfile>
```

```
RDKitGenerateMolecularFrameworks.py -h | --help | -e | --examples
```

DESCRIPTION

Generate Bemis Murcko [Ref 133] molecular frameworks for molecules. Two types of molecular frameworks can be generated: Graph or atomic frameworks. The graph molecular framework is a generic framework. The atom type, hybridization, and bond order is ignore during its generation. All atoms are set to carbon atoms and all bonds are single bonds. The atom type, hybridization, and bond order is preserved during generation of atomic molecular frameworks.

The supported input file formats are: SD (.sdf, .sd), SMILES (.smi, .csv, .tsv, .txt)

The supported output file formats are: SD (.sdf, .sd), SMILES (.smi)

OPTIONS

-e, --examples

Print examples.

-h, --help

Print this help message.

-i, --infile <infile>

Input file name.

--infileParams <Name,Value,...> [default: auto]

A comma delimited list of parameter name and value pairs for reading molecules from files. The supported parameter names for different file formats, along with their default values, are shown below:

```
SD: removeHydrogens,yes,sanitize,yes,strictParsing,yes
SMILES: smilesColumn,1,smilesNameColumn,2,smilesDelimiter,space,
        smilesTitleLine,auto,sanitize,yes
```

Possible values for smilesDelimiter: space, comma or tab.

-m, --mode <GraphFrameworks or AtomicFrameworks> [default: GraphFrameworks]

Type of molecular frameworks to generate for molecules. Possible values: GraphFrameworks or AtomicFrameworks. The graph molecular framework is a generic framework. The atom type, hybridization, and bond order is ignore during its generation. All atoms are set to carbon atoms and all bonds are single bonds. The atom type, hybridization, and bond order is preserved during the generation of atomic molecular frameworks.

-o, --outfile <outfile>

Output file name.

--outfileParams <Name,Value,...> [default: auto]

A comma delimited list of parameter name and value pairs for writing molecules to files. The supported parameter names for different file formats, along with their default values, are shown below:

```
SD: compute2DCoords,auto,forceV3000,no
SMILES: smilesDelimiter,space,smilesTitleLine,yes
```

Default value for compute2DCoords: yes for SMILES input file; no for all other file types.

--overwrite

Overwrite existing files.

- r, --removeDuplicates <yes or no> [default: no]
Remove duplicate molecular frameworks. Possible values: yes or no. The duplicate molecular frameworks are identified using canonical SMILES. The removed frameworks are written to a separate output file.
- s, --sort <yes or no> [default: no]
Sort molecular frameworks by heavy atom count. Possible values: yes or no.
- sortOrder <ascending or descending> [default: ascending]
Sorting order for molecular frameworks. Possible values: ascending or descending.
- u, --useChirality <yes or no> [default: yes]
Use stereochemistry for generation of canonical SMILES strings to identify duplicate molecular frameworks.
- w, --workingdir <dir>
Location of working directory which defaults to the current directory.

EXAMPLES

To generate graph molecular frameworks for molecules and write out a SMILES file, type:

```
% RDKitGenerateMolecularFrameworks.py -i Sample.smi -o SampleOut.smi
```

To generate graph molecular frameworks, remove duplicate frameworks for molecules and write out SD files for unique and duplicate frameworks, type:

```
% RDKitGenerateMolecularFrameworks.py -m GraphFrameworks -r yes  
-i Sample.sdf -o SampleOut.sdf
```

To generate atomic molecular frameworks, remove duplicate frameworks, sort frameworks by heavy atom count in ascending order, write out SMILES files for unique and duplicate frameworks, type:

```
% RDKitGenerateMolecularFrameworks.py -m AtomicFrameworks -r yes  
-s yes -i Sample.smi -o SampleOut.smi
```

To generate graph molecular frameworks for molecules in a CSV SMILES file, SMILES strings in column 1, name in column 2, remove duplicate frameworks, sort frameworks by heavy atom count in descending order and write out a SD file, type:

```
% RDKitGenerateMolecularFrameworks.py -m AtomicFrameworks  
--removeDuplicates yes -s yes --sortOrder descending --infileParams  
"smilesDelimiter,comma,smilesTitleLine,yes,smilesColumn,1,  
smilesNameColumn,2" --outfileParams "compute2DCoords,yes"  
-i SampleSMILES.csv -o SampleOut.sdf
```

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SEE ALSO

RDKitConvertFileFormat.py, RDKitDrawMolecules.py, RDKitSearchFunctionalGroups.py, RDKitSearchSMARTS.py

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The functionality available in this script is implemented using RDKit, an open source toolkit for cheminformatics developed by Greg Landrum.

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